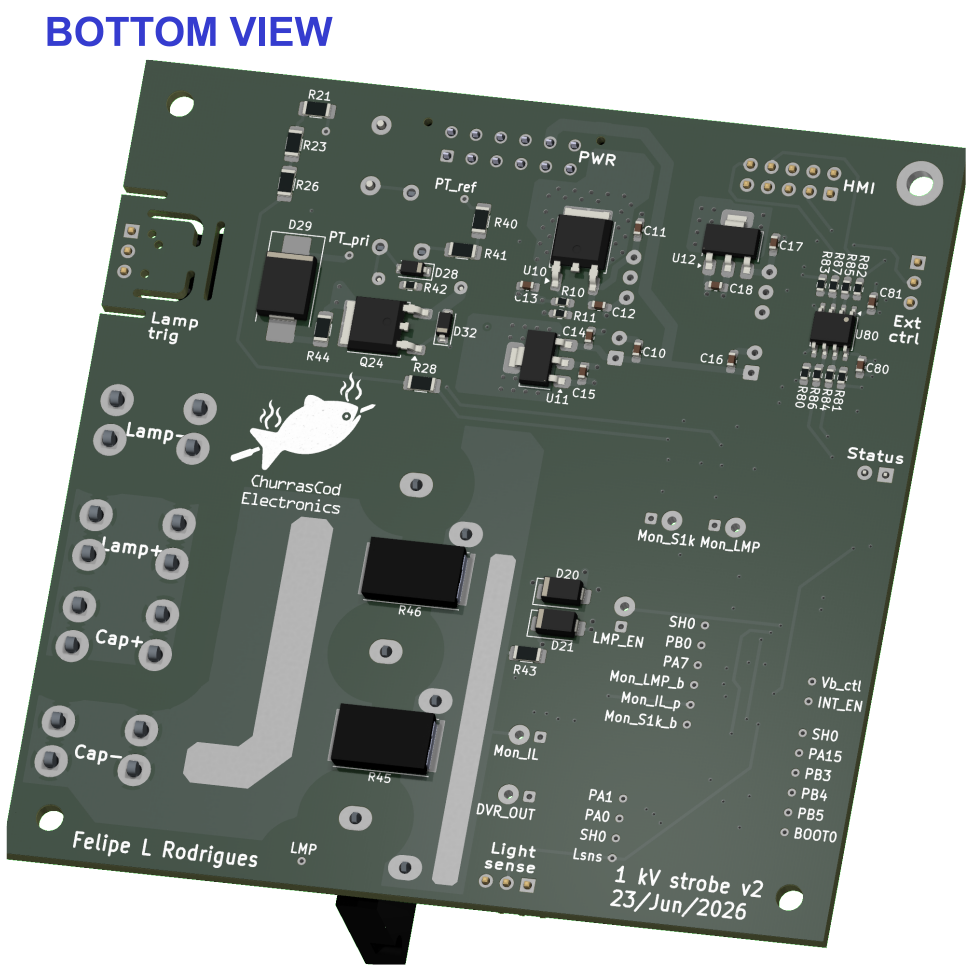
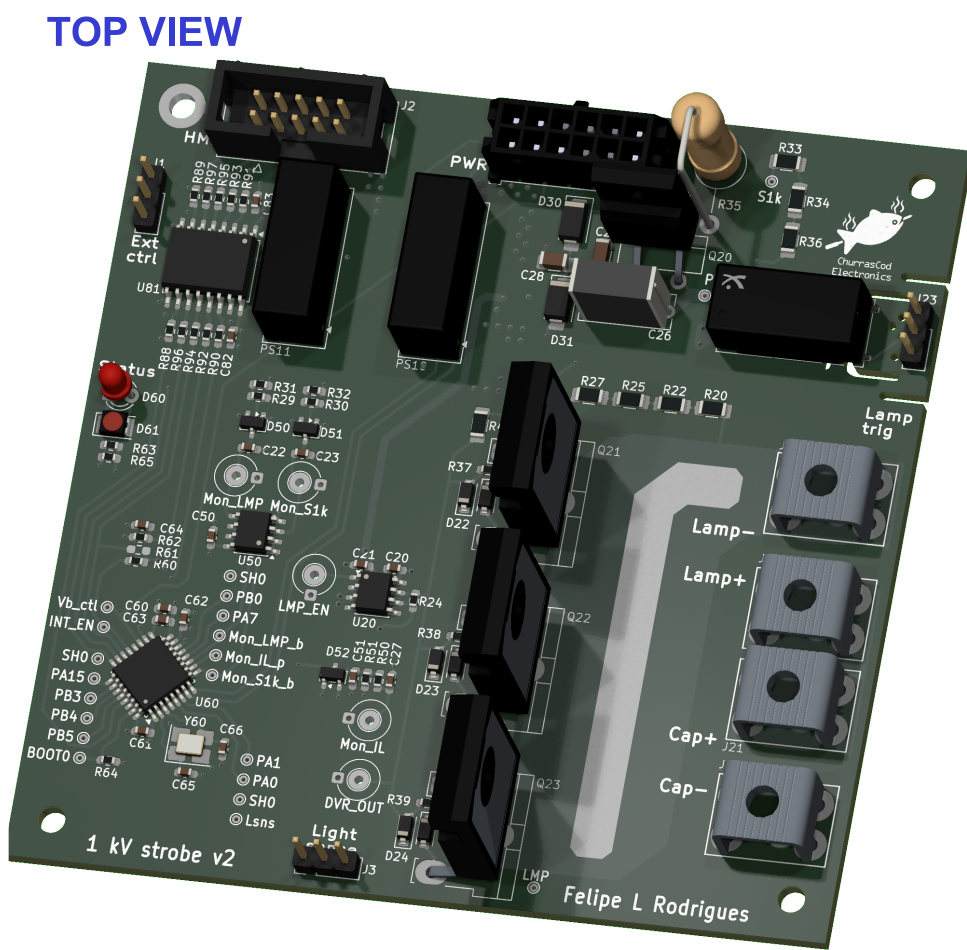


Main board

Rev 2.0.0+ (Unreleased)

2026-09-05

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1	Cover Page
2	Block Diagram
3	Power supply
4	Lamp driver
5	Signal conditioning
6	MCU
7	Optical isolation
8	Revision History

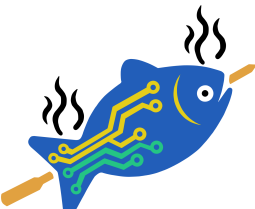


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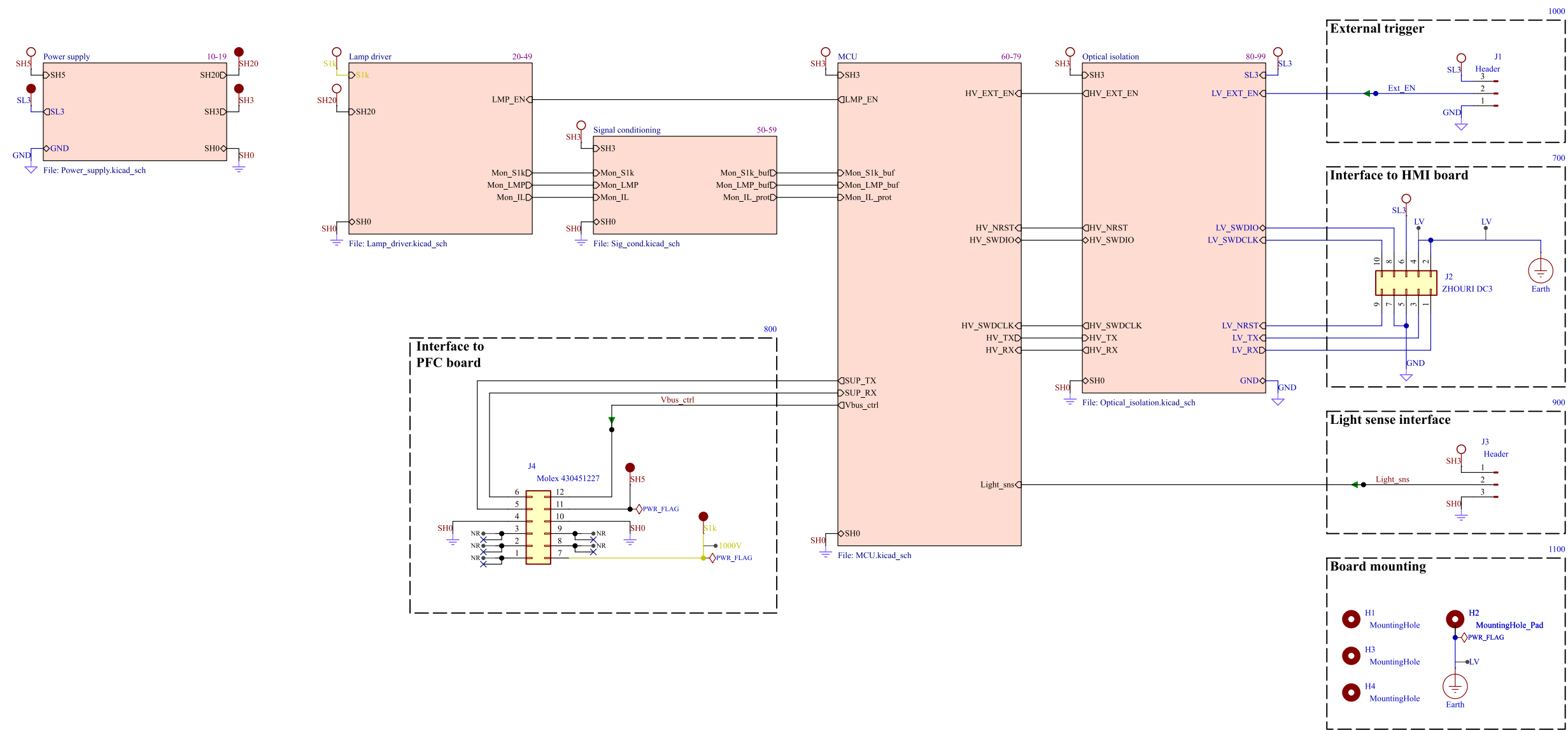
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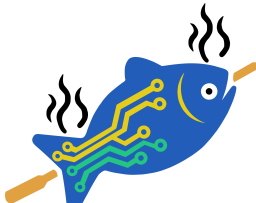
NOTE STYLING CONVENTION

DESIGN NOTE: Example text for informational design notes.	DESIGN NOTE: Example text for debug notes.	DESIGN NOTE: Example text for cautionary design notes.	DESIGN NOTE: Example text for critical design notes.	LAYOUT NOTE: Example text for critical layout guidelines.
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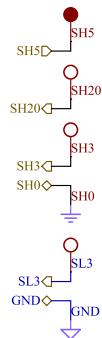
 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: 84f4bdf
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title:		File Name: Strobe_main.kicad_sch	Designer: FR	Date: 2025-07-04	Revision: 2.0.0+ (Unreleased)
	Sheet Path: /		Reviewer:		Size: A3	Sheet: 1 of 8

Block Diagram

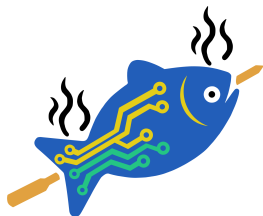
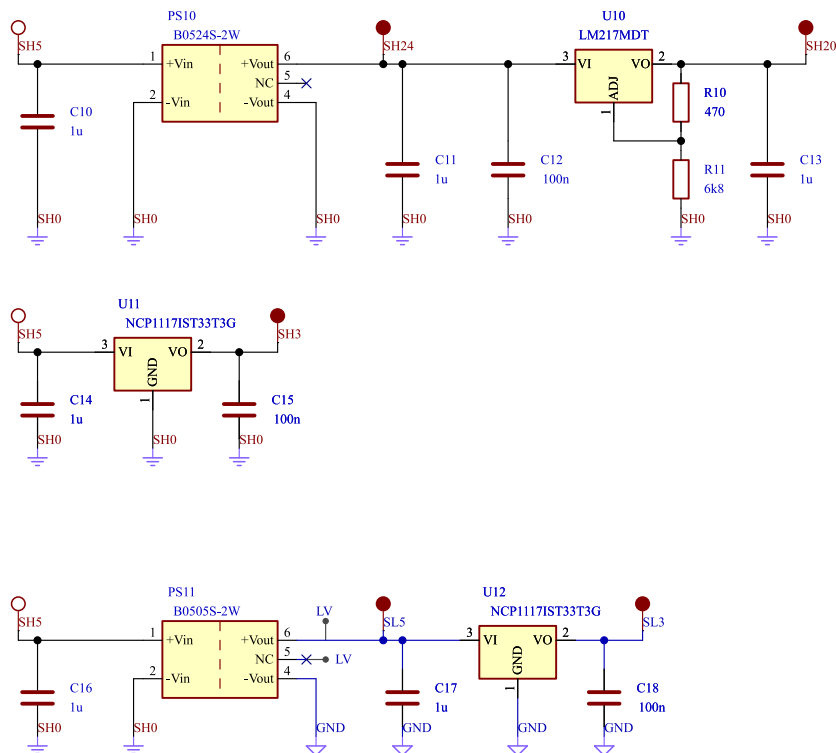


 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: CHECKED		Git Hash: 84f4bdf			
			Board Name: Main board			Project Name: 1 kV strobe				
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-07-04		Revision: 2.0.0+ (Unreleased)	
	Sheet Path: /Block Diagram/				Reviewer:		Size: A3		Sheet: 2 of 8	

BLOCK INTERFACE



Power supply 10-19



ChurrasCod
Electronics

Comments:

Sheet Title:

Power supply

Sheet Path:

/Block Diagram/Power supply/

Company:

Board Name:

Main board

File Name:

Power_supply.kicad_sch

Designer:

FR

Reviewer:

Variant:

CHECKED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

Git Hash:

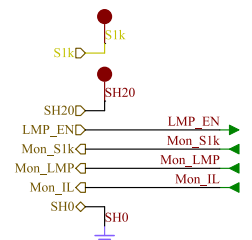
84f4bdf

Revision:

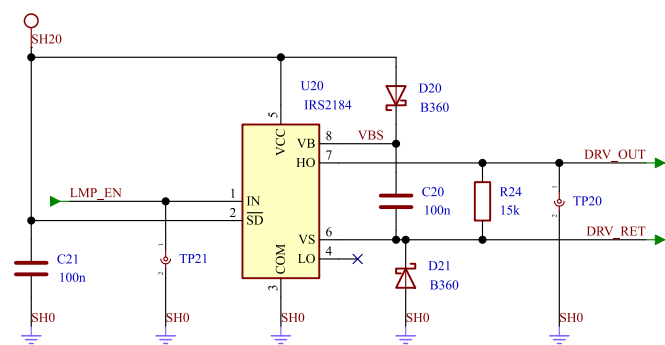
3 of 8

Lamp driver

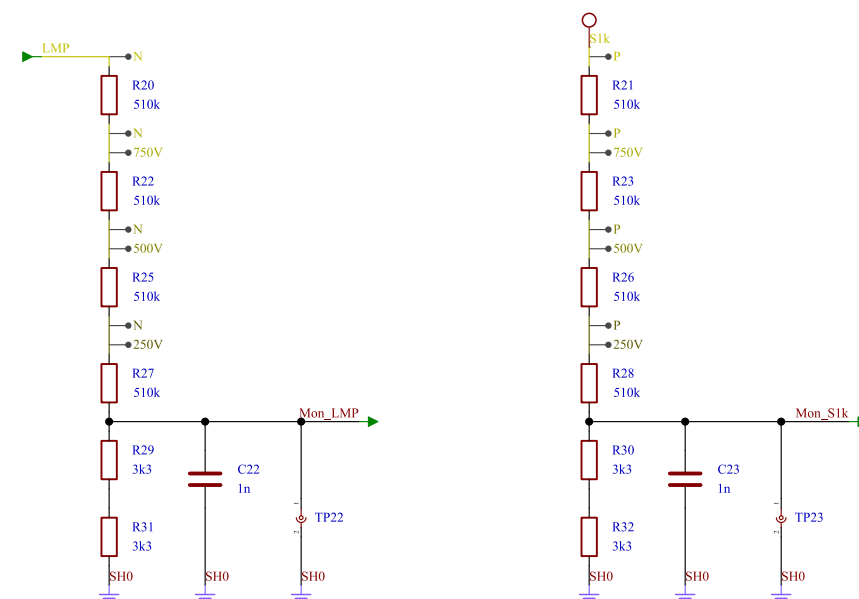
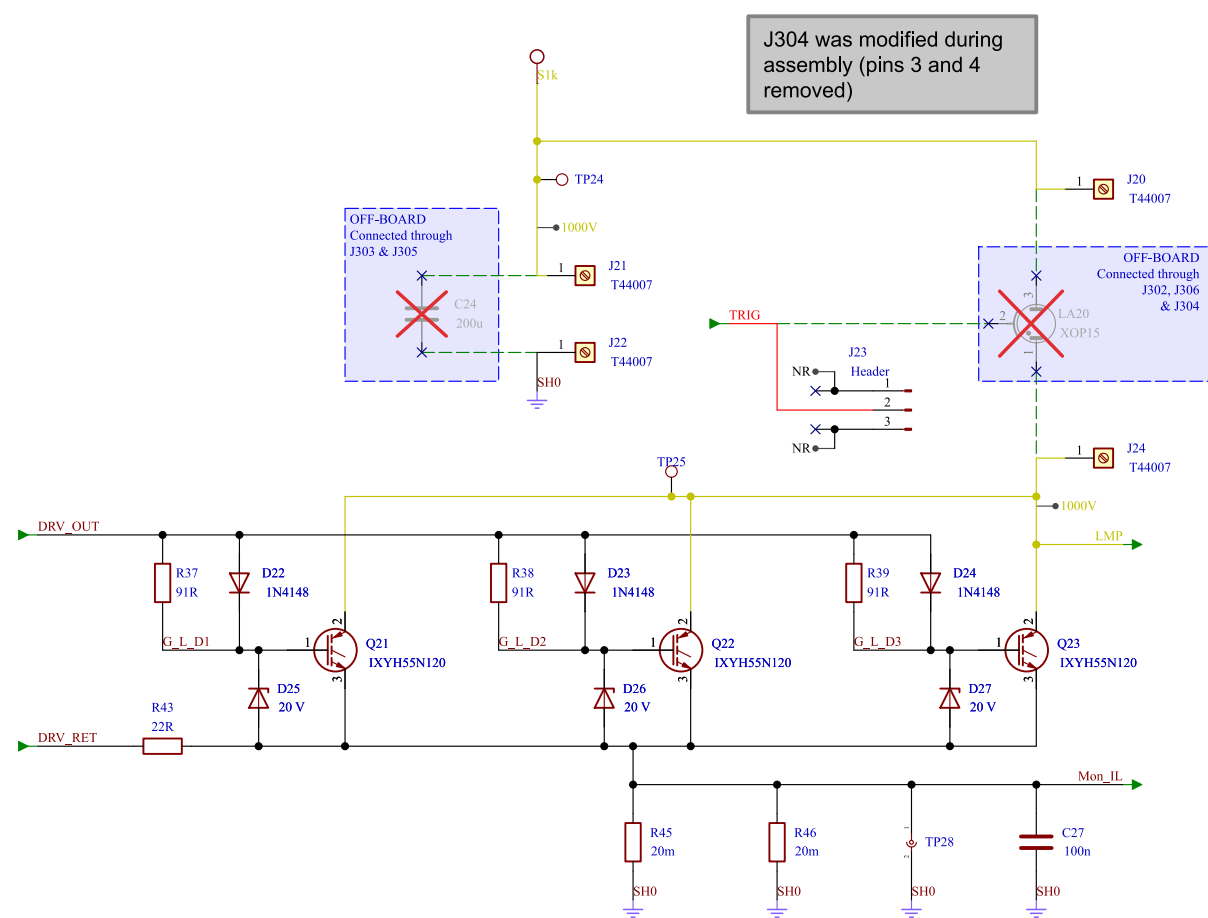
20-49



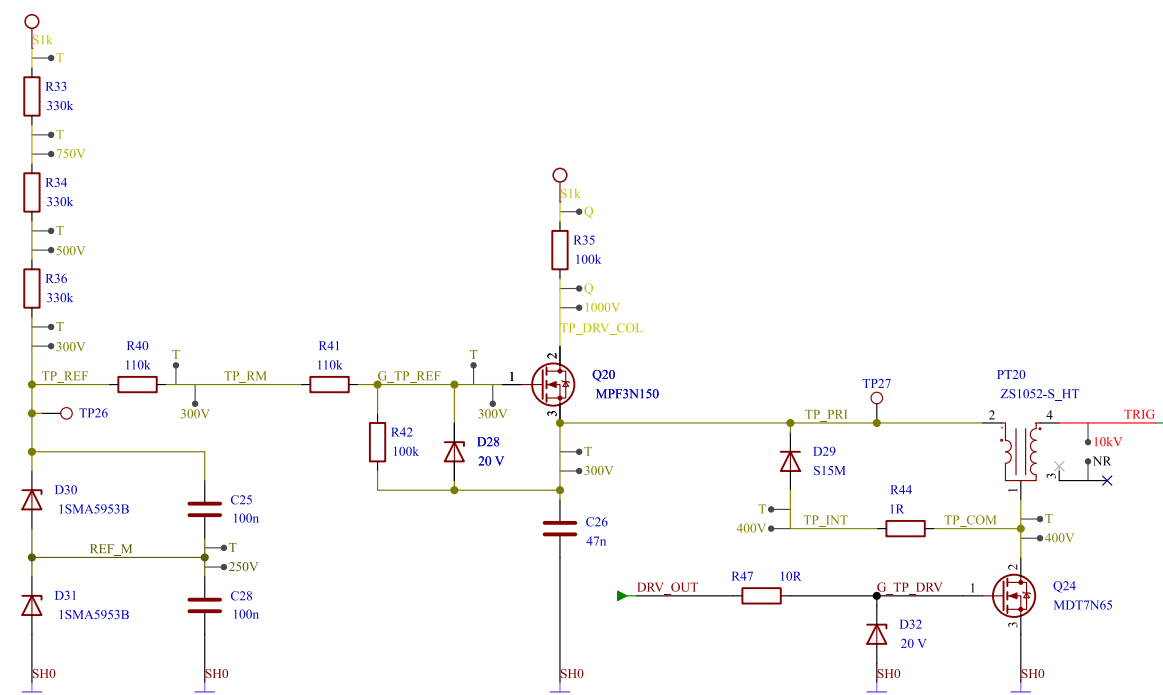
INPUT STAGE (IGBT DRIVER)



OUTPUT STAGE

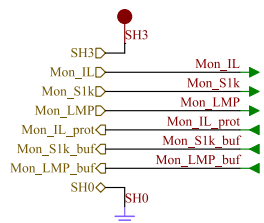


PULSE TRANSFORMER DRIVER

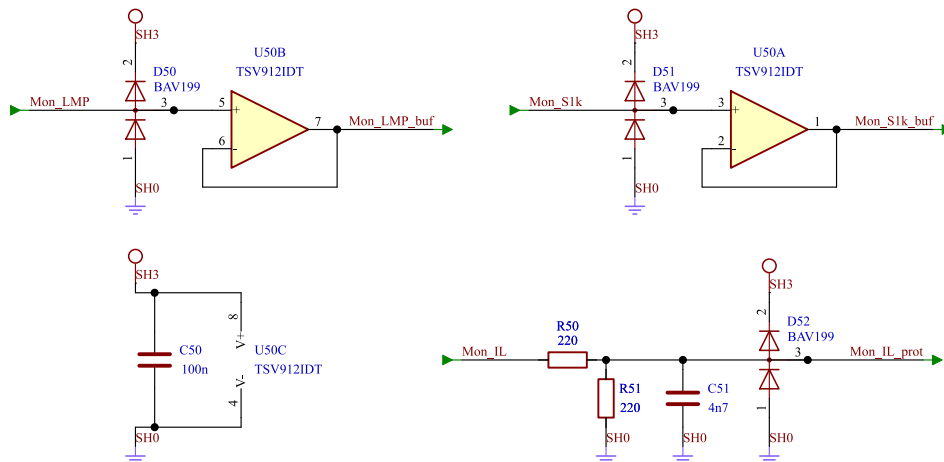


Comments:	Company:		Variant: CHECKED	Git Hash: 84f4bdf
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Lamp driver	File Name: Lamp_driver.kicad_sch	Designer: FR	Date: 2025-07-04	Revision:
Sheet Path: /Block Diagram/Lamp driver/		Reviewer:	Size: A3	Sheet: 4 of 8

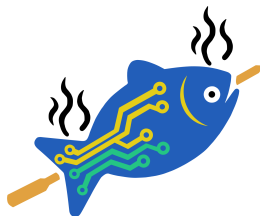
BLOCK INTERFACE



LAMP DRIVER VOLTAGES & CURRENT



Signal conditioning 50-59



ChurrasCod
Electronics

Comments:

Company:

Variant:

Git Hash:

CHECKED

84f4bdf

Board Name:

Main board

Project Name:

1 kV strobe

Sheet Title:

Signal conditioning

File Name:

Sig_cond.kicad_sch

Designer:

FR

Date:

2025-07-04

Revision:

Sheet Path:

/Block Diagram/Signal conditioning/

Reviewer:

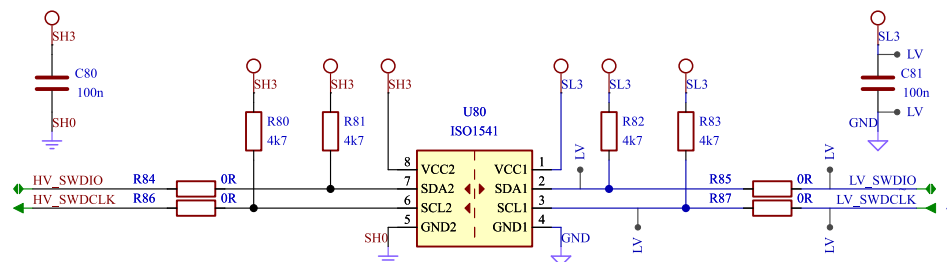
Size:

A4

Sheet:

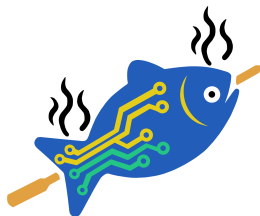
5 of **8**

Pin connection diagram for the HV module. The diagram shows a top header with pins SH3, SL3, and SH0. SH3 is connected to HV_SWDIO. SL3 is connected to HV_SWDCLK. SH0 is connected to HV_EXT_EN. The bottom header has pins GND, SH0, and SH0. SH0 is connected to HV_EXT_EN. The diagram shows connections for HV_SWDIO, HV_SWDCLK, HV_NRST, HV_TX, HV_RX, LV_TX, LV_RX, LV_SWDIO, LV_SWDCLK, LV_NRST, LV_EXT_EN, and HV_EXT_EN.



The schematic diagram illustrates the internal circuitry of the HV module, centered around the ISO6741 chip (U81). The chip is configured with its V_{cc1} and V_{cc2} pins connected to the SH3 pin through a 10k resistor (R88). The GND1 and GND2 pins are connected to the SH0 pin through a 10k resistor (R98). The chip's output pins (OUTA, OUTB, OUTC, OUTD) are connected to the HV module pins (HV_RX, HV_NRST, HV_EXT_EN, HV_TX) through 0R resistors (R90, R92, R94, R96). The input pins (INA, INB, INC, IND) are connected to the HV module pins (LV_TX, LV_NRST, LV_EXT_EN, LV_RX) through 0R resistors (R91, R93, R95, R97). The chip is also connected to the SH3 pin through a 10k resistor (R89) and to the SH0 pin through a 10k resistor (R98). The chip is connected to the SH3 pin through a 10k resistor (R89) and to the SH0 pin through a 10k resistor (R98). The chip is connected to the SH3 pin through a 10k resistor (R89) and to the SH0 pin through a 10k resistor (R98).

Optical isolation



ChurrasCod
Electronics

Comments:	Company:		Variant: CHECKED	Git Hash: 84f4bdf
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Optical isolation	File Name: Optical_isolation.kicad_sch	Designer: FR	Date: 2025-07-04	Revision:
Sheet Path: /Block Diagram/Optical isolation/		Reviewer:	Size: A4	Sheet: 7 of 8

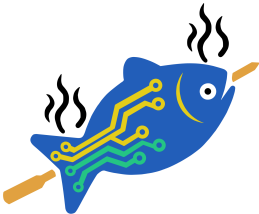
Revision History

Version 2.0.1 not found.
Version 2.0.1 not found.

- Version 2.0.0 - 2026-06-23**
- decreased impedance of Ipeak measurement divider, removed buffer and fed signal directly to MCU
 - reworked gate drive resistors
 - reworked power input
 - modified some of the testpoints and added extra "GND testpoints"
 - reworked test pulse circuit to achive higher frequency
 - added mlcc to MCU rst line

Version 1.0.0 - 2026-04-11

Migrated existing project to template (KDT)



ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: 84f4bdf
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-07-04	Revision: 2.0.0+ (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 8